



Pretend play is not an ‘extra’: Exploring the relation between parent-reported pretend play and executive functions in the toddlerhood period[☆]

Joanna Kelly¹ , Tobias Constien¹ , Michelle Downes^{*}

University College Dublin, Dublin, Ireland

ARTICLE INFO

Keywords:

Toddlers
Pretend play
Executive functions
Cognitive development
World Café

ABSTRACT

Empirical evidence and theoretical accounts increasingly highlight the role of pretend play in fostering the development of executive function skills in childhood. Yet, previous research has predominantly been focused on pre-school- and primary-school-aged children, even though both pretend play and executive functioning have a developmental onset much earlier in development. To address this gap within the existing evidence base, the current report aimed to explore this association in the toddlerhood period using a multi-method approach. In an initial consultation, insights from mothers of toddlers were obtained via a World Café approach. Secondly, a parent-reported online study, including 213 toddlers ($M = 25.47$ months, $SD = 5.33$) identified a quantitative estimate of the concurrent relation between parent-reported indices of pretend play and executive functions in toddlerhood. After controlling for demographic variables, a significant association between toddlers' pretend play ability and cognitive executive functioning skills was identified. However, no relation was found between toddlers' pretend play ability and regulatory executive function skills. Findings from across studies are discussed in relation to previous theoretical accounts on the role of pretend play in development.

Play is an almost ubiquitous phenomenon of early childhood. Many early parent-child interactions are inherently playful (e.g., peek-a-boo, nursery rhymes, shared book readings; [Tamis-LeMonda et al., 2002](#)), family routines are rehearsed and practised in play (e.g., cooking with a toy kitchen, feeding a doll; [Lillard, 2010](#)), and early education curricula capitalise on children's inherent propensity towards play ([Sutterby & Kaharod, 2020](#)). Consequently, children's play has long been recognised as fundamental by anthropologists, sociologists, and psychologists ([Whitebread et al., 2019](#)) and is formally enshrined in the [United Nations \(1989\)](#). Within the field of developmental psychology specifically, pretend play, a type of play that is imaginative and creative (e.g., talking on a toy phone, pretending to cook, etc.; [Gleason & White, 2023](#)), has been highlighted as a contributor to important developmental outcomes like executive functions ([Constien et al., 2026](#)), language ([Quinn et al., 2018](#)), or pro-social behaviour ([Smits-van der Nat et al., 2024](#)).

Theoretical accounts have long conceptualised a link between children's pretend play and their cognitive development ([Nicolopoulou, 1993](#)), specifically in regard to executive functions ([Carlson & White, 2013](#); [Constien et al., 2026](#); [Gleason & White,](#)

^{*} This study was pre-registered on AsPredicted.org (#213815). Research materials are publicly accessible on this study's OSF project page (osf.io/a5473). The authors would like to sincerely thank all the participants in this study for their time and input.

^{*} Correspondence to: School of Psychology, University College Dublin, Belfield, Dublin Ireland.

E-mail address: michelle.downes@ucd.ie (M. Downes).

¹ The first and second named author would like to be recognised as joint first authors of this work.

2023). Executive functions are key cognitive skills that allow conscious control of thoughts, actions, and emotions (Zelazo & Carlson, 2012). They have been shown to be predictive of children's school readiness and academic performance and are therefore identified as valuable targets of early interventions (Diamond, 2016). Recent empirical studies have consistently shown a relation between pretend play and executive functions in children's early development (e.g., Bauer et al., 2021; Carlson et al., 2014; van Reet, 2020). Moreover, play-based programs implemented in pre-school or primary-school settings have shown potential to enhance learning and development, resulting, among others, in developmental gains in executive functioning of preschoolers (Muir et al., 2023). Overall, evidence shows that pretend play may be a fertile context for the practice and acquisition of key executive function skills.

Most studies on the relation of pretend play and the early development of executive functions have been conducted with preschoolers (Constien et al., 2026). Yet, both pretend play and executive functioning have their developmental onset much earlier than the preschool period. For example, early instances of pretend play can be observed in toddlerhood, beginning around 18 months of age (Fleer, 2024). Similarly, infancy and toddlerhood are sensitive periods of executive functions development (Thompson & Steinbeis, 2020). Moreover, given methodological advances, these skills can be reliably assessed in toddlers as young as 24 months of age (Zelazo et al., 2024). As such, the current evidence-base on the relation of pretend play and executive functions is effectively missing the crucial early developmental stage of toddlerhood. This is critical, given that executive function interventions may be particularly fruitful when implemented early in development, to ensure the developmental success and academic readiness for all children (Diamond, 2016).

The current report aims to add to the growing evidence base on the relation between pretend play and executive functions in early childhood via a parent consultation and empirical study. By intentionally placing the emphasis on the toddlerhood period between 18 and 36 months, this report seeks to extend the current evidence base of this under-researched period of development and provide an initial exploration which may benefit future intervention research. First, a consultation with parents of toddlers via a World Café will seek to explore the role of play in children's cognitive development from the participating parents' perspective. Secondly, a quantitative parent-reported survey of toddler's pretend play and executive functions development will provide a quantitative estimate of the relation between pretend play and toddler's emerging executive function skills.

1. Background

The period of children's early development is marked by high neural plasticity and, correspondingly, ongoing changes to the structure and organisation of the brain (Fiske & Holmboe, 2019; Varghese et al., 2026). Grey matter increases volumetrically until around the age of four, after which it undergoes selective pruning, resulting in a reduction of grey matter (Lebel & Deoni, 2018). Moreover, the structural development of the prefrontal and parietal cortices, the brain networks most strongly associated with executive functioning, takes place predominantly within a child's first two years of life (Morgan et al., 2024). Corresponding to this rapid neural development are consistent improvements in children's early executive function skills and abilities, indicating that toddlerhood (i.e., 18–36 months), between late infancy and the pre-school period, may be a sensitive period for the development of executive skills (Fiske & Holmboe, 2019; Varghese et al., 2026). Foundational executive functions begin to emerge in late infancy, with infants as young as 5–8 months demonstrating an early measure of working memory by maintaining attention to an object, even when it is not visible (Bell, 2012). Early indications of inhibition develop between 8 and 12 months, in which children show the ability to suppress external stimuli that may distract them from an object (Cuevas et al., 2012).

While in adults, the predominant theoretical model of executive functions, i.e., the unity and diversity model, delineates three main components, i.e., updating (working memory), shifting (cognitive flexibility), and inhibition (Miyake et al., 2000; Miyake & Friedman, 2012), the structure and organisation of executive functions in early childhood remains debated (Doebel, 2020; Ibbotson, 2023; Zelazo & Carlson, 2023). Models developed within the context of children's development, for instance, describe the different executive function skills (e.g., working memory, cognitive flexibility, inhibitory control) to be inseparable in children's early development, and instead conceptualise early executive functioning as a unitary, rather than multidimensional, executive construct (e.g., Hughes et al., 2009; Karr et al., 2018; Miller et al., 2012). For example, Wiebe et al. (2011) showed via confirmatory factor analysis in a sample of over 200 three-year olds that a unitary model fit their data best, which was collected across seven commonly used executive function tasks. In turn, a two-factor model, which sought to distinguish between inhibition and working memory, showed no improved model fit and indicated substantial overlap between the two factors. Consequently, Wiebe et al. (2011) concluded that executive functions are best conceptualised as a unitary composite, rather than separate components, in early childhood.

Moreover, a prominent developmental model of executive functioning conceptualises a continuum from hot (i.e., emotional) to cool (i.e., cognitive) executive skills (Zelazo & Carlson, 2012). Hot executive functions describe higher-order cognitive skills that are involved in goal-directed, or emotive contexts, such as waiting for an anticipated treat or calming oneself down after small frustrations (like not getting that treat after all). Cool executive functions, on the other hand, describe higher-order cognitive skills that are involved in more abstract, decontextualised thinking, like persisting in a challenging task (e.g., completing a peg puzzle), or keeping information in mind (e.g., where did the elephant piece go again?). Importantly, within this conceptualisation, executive functioning is not strictly hot or cool, but rather a matter of extent (Zelazo et al., 2024). Ultimately, both hot and cool executive skills jointly shape children's everyday functioning (Hughes & Katus, 2024). When toddlers are told that they have to put back a new and exciting toy, for example something they see in the toy aisle while shopping, they have to control their initial impulses (e.g., holding onto the toy and refusing to give it back), manage their potential frustration, remember potential rules (e.g., "we are not buying toys today"), and accept an alternative (e.g., "we can come back another time"). Consequently, most everyday situations in children's development demand both hot and cool executive functioning skills.

1.1. The role of pretend play in children's early executive development

Within theoretical conceptualisations, pretend play has long been linked to children's early cognitive development (Berk & Meyers, 2013; Nicolopoulou, 1993). Piaget (1951) conceptualised pretend or "symbolic" (p. 111) play as an indicator of children's emergent cognitive skills in their early years of development. Moreover, Vygotsky (1967) famously stated that "a child's greatest self-control occurs in play" (p. 14), and conceptualised pretend play as actively contributing to children's early cognitive development. Specifically, in regard to executive functions, more recent theoretical conceptualisations identify a bi-directional relation between pretend play and executive functions (Carlson & White, 2013; Constien et al., 2026; Gleason & White, 2023). Skills like working memory, inhibition, or cognitive flexibility are needed for children to engage in pretend play, while at the same time, pretend play provides a fertile context to apply and practice these emergent skills (Carlson & White, 2013). Pretending to cook at a toy kitchen, for instance, requires skills of working memory (keeping roles and rules in mind, e.g., this block is a carrot), and cognitive flexibility (adapting to other's ideas, e.g., this block can also be an orange). At the same time, it allows children to practice emerging executive skills like planning and organisation (e.g., deciding what to cook and how), and persistence (e.g., staying within the cooking scenario).

Previous experimental and intervention research has shown that pretend play has the potential to strengthen children's executive functioning in the short- but also long-term (Bierman & Torres, 2016; Diamond & Ling, 2019; Muir et al., 2023). To date, the limited evidence base on the effectiveness of pretend play interventions has been mixed, indicating that further work, specifically exploring contextual factors (e.g., parental involvement, cultural factors), is needed (Doebel & Lillard, 2023). Thibodeau-Nielsen et al. (2020), for example, showed that in a daily, five-week pre-school intervention, 2–5 year-old children from a middle-class, US-background participating in a fantastical pretend play intervention showed significant gains in their cool executive functioning in contrast to children participating in realistic pretend play, non-imaginative play, and a control group. However, these effects did not extend to students from a lower-income background, indicating a potential moderating role of socio-economic status or culture (Thibodeau-Nielsen et al., 2020). In regard to observational research, a recent quantitative meta-analysis identified a small yet significant relation between pretend play and executive functioning in early childhood (i.e., 12–72 months) across 26 unique studies and nearly 3000 children (Constien et al., 2026). Yet, despite evidence that even young children engage in pretend play (e.g., Fleeer, 2024) and show evidence of emerging executive skills (Morgan et al., 2024), only a few studies included children under 36 months (i.e., toddlers), indicating a significant evidence gap.

First, Rutherford and Rogers (2003), in a sample of toddlers with and without autism, showed that scores on a task of generativity, an executive skill that allows children to generate novel ideas to explore or solve problems (e.g., when sorting shapes into a shape sorter), was linked to toddlers' level of pretend play, even after accounting for children's verbal mental age. Notably, no relation between generativity and sensorimotor play was shown, indicating a unique relation between pretend forms of play and executive functioning (Rutherford & Rogers, 2003). Similarly, Carlson et al. (2004) identified a significant relation between a comprehension of pretense task and an executive functioning composite score, which included hot and cool tasks of executive functioning, in 24-month-old toddlers, even after controlling for children's age, sex, and verbal ability.

Conversely, Yeung et al. (2019), in a sample of 20–31-month-old toddlers, identified no bi-variate correlations between scores on a similar task of pretend play comprehension and toddler's executive skills, as assessed via a series of relatively cool executive function tasks. Lastly, van Reet (2020), utilising a similar measure of pretend play comprehension, showed toddlers' delay inhibitory control, assessed via a relatively hot delay of gratification task with stickers, to be significantly related to their scores on the pretence task. In turn, the two other tasks of conflict inhibitory control and working memory, two relatively cool aspects of executive functioning, showed no relation to toddlers' comprehension of pretense (van Reet, 2020).

Overall, previous literature on intervention and cross-sectional research shows mixed findings in regard to the developmental role of pretend play in early childhood. While there is increasing evidence indicating a relation between measures of pretend play and executive functioning (Constien et al., 2026), as well as a strong theoretical link between the two domains (Berk & Meyers, 2013; Gleason & White, 2023; Nicolopoulou, 1993), intervention research does not clearly show a strong causal role of pretend play on children's early executive development (Lillard et al., 2013; Doebel & Lillard, 2023). Consequently, aside from a call for further research, specifically intervention and longitudinal studies, further investigation into the contextual factors impacting children's pretend play and executive functions to further explore their relation has been called for (Constien et al., 2026; Doebel & Lillard, 2023). One particular contextual factor that may be particularly important in earlier periods of development such as toddlerhood may be parental attitudes towards play.

1.2. Parental attitudes towards play and its developmental role

Specifically in children's early development, parents, a child's first playmates, substantially shape how and how often their child engages in play (LaForett & Mendez, 2017). On the one hand, parents are providing toys as well as the necessary space at home, which can determine how their child plays. Open-ended toys, like stuffed animals or dolls for example, may elicit more pretend play, whereas puzzles or toy blocks can potentially encourage more structured play (e.g., Creaghe et al., 2021). On the other hand, parental involvement is also instrumental in guiding and scaffolding their child's play (Lillard, 2010). Dunn and Wooding (1977), for example, showed that 18–24 months-old toddlers were more likely to engage in pretend play when they were playing with their mother, than when playing by themselves. Effectively, parental attitudes towards play can largely influence children's play both in terms of quantity and quality (Lillard, 2010). In a series of cross-cultural studies of Ethiopian and Hungarian preschoolers (i.e., 4–6 years), Metaferia et al. (2021) showed that positive parental attitudes towards play as a developmentally important activity were positively linked to the frequency of their children's pretend play, as well as the development of cool executive functions, specifically their inhibitory

control.

Specifically within Ireland, the Irish Research Council commissioned a national survey to examine the perspectives of parents of pre-schoolers (i.e., 4–6 years) on school readiness (Ring et al., 2016; Ring & O'Sullivan, 2017). Word frequency analysis of this qualitative data found that over 60% of parents mentioned the importance of academic skills, such as the ability to hold a pencil and knowledge of letters and sounds. However, only 13% of parents specifically referred to the role of play in supporting development as well as emergent academic, social, and emotional readiness, suggesting that parents perceive a divide between play and learning and that, particularly when it comes to school-readiness, parents prefer an emphasis on the latter (also see Kane, 2016). More recent research internationally, however, shows shifting attitudes from parents towards play. Wright et al. (2023), for instance, showed in a nationally representative US-sample of parents of children aged 2–12 years largely positive attitudes towards play-based learning. Parents, for instance, were most likely to concur that learning results from free play (Wright et al., 2023). Similarly, Nesbit et al. (2025) showed in a recent national survey of Britain, that a majority of parents of primary school children perceive playtime as a vital part of their child's school day.

2. Parent consultation

Specifically, within the early years (i.e., before children enter pre-school), parents play a significant role in shaping and scaffolding their child's play (Lillard, 2010). Effectively, their attitudes towards play largely influence the quality and quantity of play (Metaferia et al., 2021). Most research to date, however, has focused on parents of preschool-aged children. Therefore, further research is needed to provide insight into parental perspectives on the role of play in earlier periods of development, such as toddlerhood. Consequently, this preliminary consultation was comprised of a World Café that aimed to explore perceptions and experiences of play from parents of toddlers in Ireland. The World Café method is an inclusive, participatory qualitative research methodology that invites participants to a shared discussion on a specific topic at round tables to generate novel insights, ideas, or solutions (Löhr et al., 2020; Schiele et al., 2022). Within a World Café, participants are recognised as experts based on their lived experience and knowledge of the topic of discussion (Recchia et al., 2022). Moreover, through open dialogue, guided by previously agreed upon rules of discussions, participants can build on each other's responses and thus arrive at a more comprehensive perspective of the topic at hand. This approach has been used previously specifically with parents in regards to their experience of their children's diagnosis of cerebral palsy (Byrne et al., 2019), their transition into school (Baker et al., 2019), or mind-wandering in schools (Flynn et al., 2025). Within the context of this research project, the World Café methodology was utilised to consult with parents of toddlers. By using a World Café, this project sought to arrive at a more comprehensive understanding of parents' perspectives and experiences of their child's play during the toddlerhood period, to complement the quantitative data present in Study 2. The question guiding the discussion at the World Café was as follows:

RQ. What are parents' perceptions and experiences of toddlers' pretend play and its role in children's cognitive development?

3. Method

3.1. Design

This parent consultation utilised an in-person World Café research event. The World Café method employs an informal, community-based approach in which participants gather in a relaxed, inclusive, and participatory environment to discuss key questions (Brown & Isaacs, 2005; Löhr et al., 2020; Schiele et al., 2022). Prior to the World Café event, ethical approval was ensured from University College Dublin (TMREC-PSY: 2024-28).

3.2. Participants

Inclusion criteria identified eligible participants as parents or guardians of toddlers between the ages of 24 and 36 months. Participants were invited to take part in the study through a variety of recruitment methods (e.g., local libraries, toddler groups, social media). Eleven participants registered to attend, however only five participants attended the in-person World Café event. All participants ($N = 5$, 100%) were mothers, who identified as White. Additionally, two participants (40%) identified as stay-at-home parents, while the other three were working full-time. All participants had completed a third-level university degree.

3.3. Procedure

Participants registered for the event online, where they were also provided with an information sheet and consent form. The event was held on February 17th, 2025 and lasted 90 min in total. At the beginning of the event, the concept of play was introduced to participants as a common everyday experience, but one which is often difficult to define, and thus requires further exploration. The room consisted of two round tables, each with World Café tablecloths, pens, sticky notes, and World Café etiquette cards (e.g. "Direct your focus on what is important", "Speak and listen with heart in mind", etc.). Each table, consisting of a moderator and a group of participants, had a specific question displayed on it. Participants were split up between the two tables (i.e., groups of two and three). To begin, the moderators introduced themselves to the group and explained their roles. The moderator at each table began with an icebreaker (i.e., sharing your favourite childhood game) to create a warm, inviting environment for the participants. Moreover,

participants were invited to have tea, coffee and snacks throughout the event. The two key questions guiding the discussion at the two tables were:

Table 1) What does play look like in toddlerhood, and what is its role in development?

Table 2) What are your experiences of your toddler engaging in pretend play?

Participants then spent 15 min at each table, engaging in conversation around the question assigned to that table, before moving to the following table. With the new group, the moderator outlined the question and summarised the discussions generated by the previous group. Participants were encouraged to read previous groups' thoughts and record their ideas on the paper tablecloths as their conversations developed (see Fig. 1). Additionally, the moderators took notes of participants' conversation which at times were verbatim quotes (e.g., "show off", "learning through doing") but also included summative statements (e.g., "picks up things from older children"). No audio was recorded during the event to support the informal nature of the World Café approach. At the end of the session, a harvest round took place, consisting of a group discussion to capture the unique insights from conversations in the room. Each moderator read aloud their table's thoughts and findings, which were then summarised by the facilitator. Participants were also given the opportunity to add any additional thoughts or ideas they felt might have been missed to ensure their thoughts were accurately represented and understood by the moderators.

3.4. Data analysis

Following the completion of the World Café, the two moderators digitally transcribed the tablecloths and the harvest round posters, and any discrepancies were resolved in discussion with a third party. Qualitative data were analysed using Braun and Clarke's (2022) method of reflexive thematic analysis (RTA), a qualitative method of analysis that seeks to develop, analyse, and interpret patterns across qualitative datasets. The RTA method was chosen because it acknowledges the importance of researchers' subjectivity (Braun & Clarke, 2021), given that the researchers, as moderators, were part of the discussion at the World Café. The first author initially identified codes and created themes using the six-step procedure proposed by Braun and Clarke (2022). Following this provisional analysis, the first and second authors reviewed and identified themes, and collaboratively finalised the analysis and its reporting. To illustrate, the second theme was originally named "The Social Aspect" and subsequently renamed "Co-construction of play" to further highlight the active role of children and parents in play following the authors' discussion. For the purpose of reporting, notes taken during the World Café (see Fig. 1) were used to illustrate the identified themes.

4. Findings

Three themes were generated from notes taken during the parent consultation event, namely (1) Development of and through play during toddlerhood, (2) Co-construction of toddlers' play, and (3) Creating and missing opportunities for play. Themes are described in detail with reference to notes from the World Café event below, and are also visually presented in Fig. 2.

4.1. Development of and through play during toddlerhood

When discussing the role of pretend play in toddlers' development, participating mothers emphasised that toddlerhood is characterised by rapid development. They observed that pretend play "changes rapidly" during this time, evolving from "beginning simple"

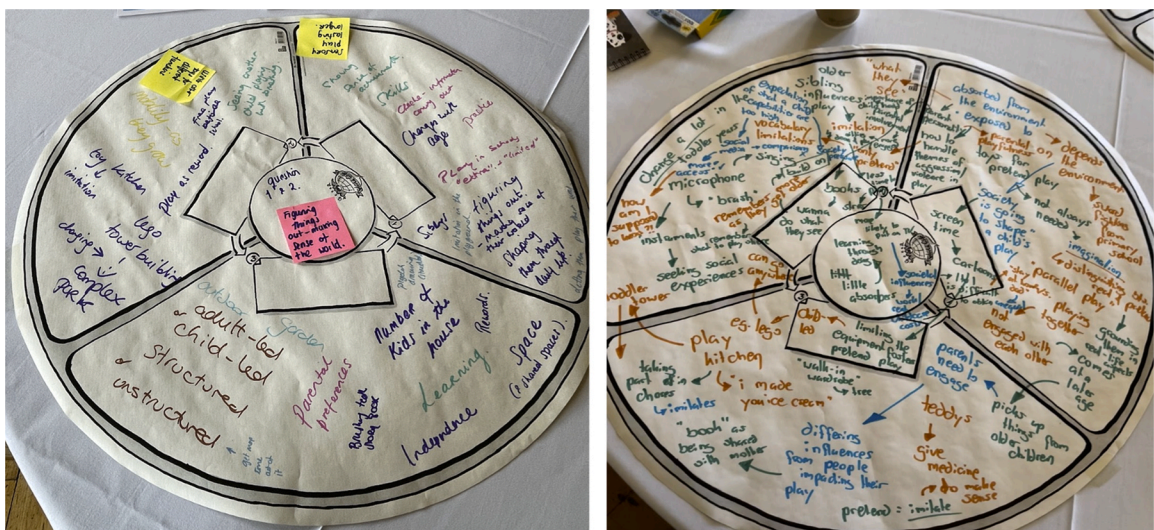


Fig. 1. Completed World Café table cloths following data collection. Note. Additional photos of materials from the World Café (i.e., set-up, harvest round) are provided in the [supplementary material](#).

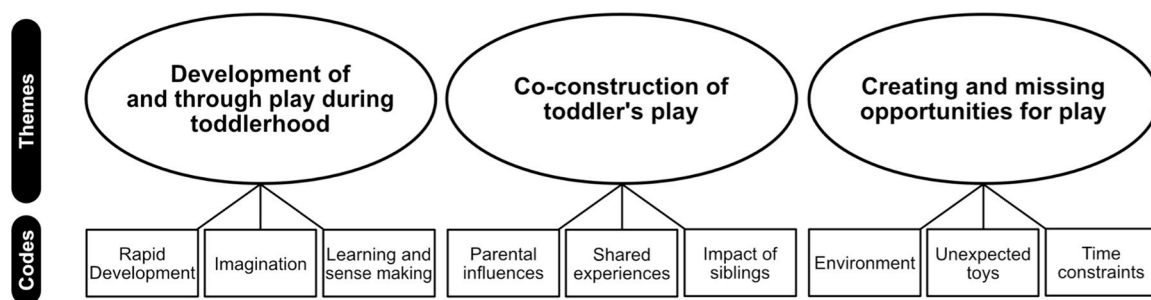


Fig. 2. *World Café: Thematic Map.* Note. The guiding research question guiding the analysis was: What are parents' perceptions and experiences of toddlers' pretend play and its role in children's cognitive development?.

to "becoming more complex". Participating mothers illustrated how, as toddlers develop, they can "remember more" and "manipulate objects" with greater skill, benefitting from increased mobility and verbal abilities, which enhances both the richness and complexity of their imaginative play. At the same time, mothers emphasised pretend play shaping development during this period. They described utilising and practising their imagination during play. This includes "using toys in alternative ways", engaging in role-playing (such as doctors) and imitating parental activities (such as chores). Participating mothers felt that engaging in pretend play afforded their toddlers seminal opportunities for learning and making sense of their world. As toddlers engaged in pretend play, it "grounded them in real-life concepts", enabling them to "learn" and "practice skills". As toddlers increasingly sought independence, there was an appreciation of "letting them play their way" as they began to "figure things out" and "make sense of the world". Ultimately, the participating mothers perceived that pretend play depends on gains in toddlers' cognitive and motor development. However, at the same time, they recognised pretend play as benefiting development, acknowledging it as an essential learning tool for toddlers, practising developing skills and allowing them to make sense of their everyday experiences during this period of rapid development.

4.2. Co-construction of toddler's play

A second theme that emerged was the social aspect of pretend play, centring on the "different influences from people impacting their play" and the understanding that "society is going to shape a child's play". The participating mothers noted their influential role in shaping pretend play, recognising how factors such as "parent personality" and "parental preferences" can influence it. Consequently, they emphasised the "need" for parents to engage in the "back and forth" pretend play with their toddlers "to get to the next level". However, they also emphasised that "child-led" and "unstructured" play often resulted in the most fruitful and extended pretend play sessions. These "shared social experiences" were seen as "more fun than being alone" for toddlers, offering a valuable sense of connection and a shared "sense of achievement together". Additionally, participating mothers recognised siblings as a significant social influence in pretend play, noting that toddlers are often shaped by their older siblings' play and factors such as "the number of kids in the house" and "sibling position".

4.3. Creating and missing opportunities for play

The participating mothers identified that pretend play experiences in toddlerhood often "depend on the environment", specifically the available "space" within the household or outdoor environments. They expressed that space is important for "exploration" and "messing around", which they felt supported more engaged and extended pretend play. Moreover, while the participating mothers acknowledged that "new things" can serve as "catalysts for new play", they also highlighted the value of simplicity, suggesting that "limiting the equipment fosters pretend play" and that "toys are not always needed for pretend play". Instead, toddlers often use everyday items, such as "IKEA boxes", and transform them through imagination, demonstrating that imagination can also flourish with minimal resources. However, participating mothers also reflected on the realities of daily life, acknowledging that they may be "more likely to put on the TV", particularly with busy schedules. In this context, pretend play is often "limited or thought of as an extra." Consequently, despite recognising its value, the participating mothers felt that the practical constraints of everyday life could act as a barrier to pretend play, resulting in missed opportunities for shared play.

5. Preliminary Discussion

This preliminary exploration of toddlers' pretend play and its role in children's cognitive development consulted mothers to discuss their experiences of their toddler's pretend play and explore their perspective on the role of play in development in a relaxed, café-like, communal setting. Results from the analysed notes taken during the event indicate that the participating mothers valued play in general, and pretend play specifically, as a fundamental and important activity for their toddler's cognitive development. In line with developmental theories (e.g., Piaget, 1951; Vygotsky, 1967), mothers emphasised the importance of pretend play in enabling toddlers to learn, construct and apply emerging skills, as well as to make sense of the world around them. At the same time, they noted that emerging cognitive skills and abilities enable their toddler to engage in increasingly complex pretend play, as evidenced by their

alternative use of toys (i.e. cognitive flexibility), role-playing (i.e. inhibitory control), and imitating parental activities (i.e. working memory). Overall, this indicates that the participating mothers perceived a bi-directional relationship between toddlers' pretend play and their cognitive development, where one feeds into the other. As toddlers' cognitive skills develop, they can engage in increasingly complex pretend play. As they engage in more pretend play, they can practice and refine their emerging cognitive skills.

Moreover, the participating mothers noted important contextual factors that may impact toddlers' play and, in turn, their developmental benefits. First, at a micro-level, mothers noted social aspects, such as parental influences (e.g., attitudes, engagement) but also siblings, in shaping their toddler's engagement and enjoyment of pretend play. Previous studies have explored parental playfulness, i.e., the level of engagement and creativity parents show when playing with their children, and its role in children's early cognitive development. [Cabrera et al. \(2017\)](#), for instance, showed that mothers' playfulness, assessed via a free play session with their two-year-old toddlers, significantly predicted their children's emotion regulation three years later. Reports from mothers from this World Café echo these findings, emphasising how parents can shape and scaffold their toddler's play.

Secondly, at a macro-level, the participating mothers noted how socio-environmental factors may have an impact on creating or missing opportunities for play. On the one hand, they noted that play requires appropriate spaces, at home or in outdoor settings (e.g., playgrounds), which may not always be available and thus can limit toddlers' engagement in play. On the other hand, they noted limited time as a potential barrier to finding opportunities for shared play. Overall, these findings indicate that parental attitudes, family dynamics (e.g., siblings, time for play), and socioeconomic disparities are important contextual factors to consider within the relation of toddlers' play and their cognitive development.

6. Quantitative study

After conducting the qualitative study, we determined the next step was to build on these preliminary findings and to quantify the relation between pretend play and toddlers' cognitive development, specifically their executive function skills. There is a strong theoretical background linking pretend play and children's early cognitive development, specifically the development of their executive function skills ([Gleason & White, 2023](#); [Vygotsky, 1967](#)). Although this relation has been studied previously, few studies have focused on toddlers under the age of three years ([Constien et al., 2026](#)). This gap within previous research may be due to previously identified difficulties in assessing executive functions, or pretend play within this age-group, as there are only a limited number of previously validated tools ([Escobar-Ruiz et al., 2023](#); [Hoicka & Prouten, 2024](#); [Thompson & Goldstein, 2019](#)). Moreover, research with this age group is particularly challenged due to difficulties in recruiting participants for a (sufficiently powered) lab-based study ([Doebl & Frank, 2023](#)).

Therefore, the present study sought to explore the relation between pretend play and executive functions in toddlers via previously validated parent-reported measures of executive functions, which allows online methods of data collection and overcomes difficulties in lab-based studies and recruitment. Based on previous literature and parental reports from our World Café, we hypothesised that parent-reported indices of pretend play would be positively associated with parent-reported indices of toddlers' cognitive and regulatory skills of executive functions.

Table 1
Demographic data for the included sample.

Variable	<i>n</i>	%
Toddlers' Age		
<i>M (SD)</i>	25.47 months (5.33)	
Range	18–36 months	
Toddlers' Gender		
Female	114	54%
Male	96	46%
Missing/Other	3	< 1%
Ethnicity		
White Irish	187	90%
Any other White background	12	6%
Any other background	10	4%
Number of Siblings		
Only child	86	41%
One sibling	93	45%
Two siblings	26	12%
Three or more siblings	4	2%
Parental Education		
Secondary Level Education	16	8%
Undergraduate Degree	60	28%
Postgraduate Degree	136	64%
Relationship		
Mother	205	99%
Other (including Father/Guardian)	3	1%

Note. *N* = 213. There were missing responses for siblings (*n* = 4), ethnicity (*n* = 3), relationship (*n* = 5), and education (*n* = 1).

7. Method

7.1. Design

This study employed a correlational design and collected data via an online survey. It was embedded within a larger project on toddlers' cognitive development as well as sleep, cartoons and everyday play (i.e., SCOOT). Data was collected from parents and caregivers across the island of Ireland between February 21st and March 25th, 2025. Ethical approval was obtained from the research ethics board of University College Dublin (TMREC-PSY: 2024-28), and a protocol was pre-registered (#213815, aspredicted.org/ggdg-5q39.pdf) prior to any data collection. Research materials, including the dataset and analysis script, are provided on this project's OSF project page (osf.io/a5473).

7.2. Participants and respondents

Participants were identified as toddlers between the ages of 18–36 months, living on the island of Ireland (i.e., Republic of Ireland, Northern Ireland). No other inclusion or exclusion criteria were defined. The respondents to the online survey were identified as toddlers' primary caregivers, including their mother, father, or legal guardian. Overall, 542 respondents started the survey, however, only 407 agreed to the necessary consent items and suited the inclusion criteria. In accordance with recommendations from the developers of the key outcome measure (i.e., EEFAQ; Hendry & Holmboe, 2021), a 70% completion rule was applied to ensure adequate data quality. Ultimately, 213 valid responses to the online survey remained for further analysis (see [Supplementary Figure 1](#)). Respondents who did not complete the necessary items did not differ on demographic variables from respondents who completed the survey (see [Supplementary Table 1](#)). Respondents included in the final sample for analysis came from 26 out of the 32 counties on the island of Ireland. Additional demographic data on the final sample of both toddlers and caregivers are presented in [Table 1](#).

7.3. Measures

7.3.1. Early executive functioning questionnaire (EEFAQ)

The EEFAQ is a 39-item parent-report questionnaire which assesses cognitive and regulatory executive functions in infants and toddlers aged between 9 and 30 months (Hendry & Holmboe, 2021). Respondents are asked to report their toddler's executive functioning as shown in their everyday behaviour (e.g., *stop reaching completely for something when you said "no/don't touch" or similar*) within the last fourteen days via a 7-point Likert scale ranging from *Never* to *Always*. The EEFAQ comprises a total cognitive executive functioning scale that can be further divided into three subscales: inhibitory control, working memory, and cognitive flexibility. In addition, eight items within the questionnaire comprise an executive functioning regulation score, which identifies children's executive skills in emotionally-charged situations (e.g., calming down after small frustrations, showing anger when they cannot do something, etc.). The EEFAQ is sensitive to developmental change, shows limited floor and ceiling effects, good internal consistency and short-term stability (Hendry & Holmboe, 2021).

In the current study, the EEFAQ showed good reliability for the two total scores of cognitive executive functions (i.e., CEF, $\alpha = .80$), and regulatory executive functions (i.e., RG, $\alpha = .85$). In turn, the reliability for the subscales of inhibitory control ($\alpha = .63$), cognitive flexibility ($\alpha = .65$), and specifically working memory ($\alpha = .59$) were lower. Considering this, and the potential of inflating the type one error (i.e., false positive) via multiple tests, it was chosen to not consider the subscales further in the analysis. Moreover, given that the EEFAQ was originally developed and validated in a sample of infants and toddlers up to 30 months (in contrast to our current sample which ranges up to 36 months), sensitivity analyses restricted to participants between 18 and 30 months were conducted. Given that these analyses indicated no divergent results in contrast to analyses conducted on the full sample, these analyses are reported in the [supplementary materials](#) (see [Supplementary Table 3 and 4](#)) rather than the main manuscript.

7.3.2. Early pretending survey (EPS)

The EPS is an 18-item parent-report questionnaire that aims to assess children's pretend play ability, as observed by their parents in their everyday interactions with their child (Hoicka & Prouten, 2024). In this measure, parents tick yes if their child has shown a certain pretend play action previously (e.g., *gesturing with an object, like for example pretending to brush teeth with a finger*). The EPS has demonstrated good internal reliability, construct validity, and longitudinal stability, and was developed for children between the ages of 4 and 47 months (Hoicka & Prouten, 2024). In the current study, the EPS showed strong internal consistency ($\alpha = .85$).

7.4. Procedure

The online survey was created using the online tool Pavlovica from the Open Science Tools (pavlovica.org). Respondents were invited to take part in the study through a variety of recruitment methods, including posters, social media posts, and parent and toddler groups. Upon accessing the online study, participants were presented with an information sheet and a consent form, which respondents had to read and electronically agree to before commencing the study. They were then presented with the full survey, which contained 142 questions in total. The measures used in the present study (i.e., EEFAQ, EPS) were presented first, following the demographic questions. After completing the full survey, respondents were given a unique link to enter their responses to the three game-based items from the EEFAQ at a later point. Additionally, they were provided the option to enter a raffle for a 30€ retail voucher. Overall, the survey took approximately 20–30 min to complete.

7.5. Analysis

Data analysis was conducted using R within R-Studio. Initially, composite scores for the two measures were computed according to the recommendations of the scale developers. Key variables were assessed for parametric assumptions of normality using the Shapiro-Wilk test and histograms. As all variables showed a skewed distribution, violating assumptions of normality, Spearman's Rank correlation was used to compute bivariate associations within the dataset. To test the posed hypotheses, a multiple linear regression analysis, controlling for demographic variables (i.e., age, gender, parental education), was performed to assess the impact of toddlers' pretend play ability on their executive functioning. The variables were standardised and further assumption checks (i.e., variation inflation factor, linearity, Cook's distance) were performed. Responses that included missing data were excluded during the regression analyses via listwise deletion. Analytical *N* are reported along each analysis. The full analysis script and dataset is provided on this project's OSF page (osf.io/a5473).

8. Results

8.1. Descriptive statistics

Descriptive statistics for the key variables of interest, i.e., toddlers' pretend play ability and executive functioning, are presented in Table 2. Notably, there were no significant differences of gender on key outcome measures of pretend play ability ($t(172.89) = 1.26$, $p = .21$), cognitive executive functioning ($t(208) = 1.29$, $p = .19$), or regulatory executive functioning ($t(208) = 1.14$, $p = .25$).

Moreover, correlations between key variables of interest, including demographic variables, are presented in Table 3. As both pretend play ability and executive functioning showed significant correlations with toddlers' age, partial correlations, controlling for age, are presented alongside the bivariate correlations in Table 3. Results indicated a moderate, positive correlation between toddlers' cognitive executive function and pretend play ability, which held when controlling for age, $r_s(211) = .31$, $p < .001$. Notably, the correlation between pretend play and executive functioning regulation, after controlling for age, was not statistically significant, $r_s(211) = .01$, $p = .90$.

8.2. Predictors of toddlers' executive functioning

A multiple linear regression analysis was conducted to examine whether pretend play ability positively predicted toddlers' cognitive executive functioning (i.e., model 1), after controlling for demographic variables (i.e., child age, gender, parental education). The overall model was statistically significant, $F(4, 205) = 15.29$, $p < .001$, and explained 21% of the variance in cognitive executive functioning scores ($R^2 = .23$, Adjusted $R^2 = .21$). Pretend play ability was a significant predictor of cognitive executive functioning ($\beta = .42$, 95% CI [.29, .55]). Similarly, parental education was a significant predictor of cognitive executive functioning ($\beta = .16$, 95% CI [.03, .29]). However, neither gender nor toddlers' age significantly predicted toddlers' cognitive executive functioning (see Table 4).

A second multiple linear regression analysis was conducted to predict toddlers' executive functioning regulation (i.e., model 2). The overall model was not statistically significant, $F(4, 205) = 2.19$, $p = .07$. Moreover, pretend play ability was not a significant predictor of toddlers' executive functioning regulation (see Table 4).

Lastly, given that mothers specifically remarked on the positive impact of siblings on toddlers' pretend play in our parent consultation, an exploratory analysis was conducted to examine the role of siblings in the development of toddlers' pretend play (see Supplementary Table 2). Having a sibling was a significant predictor of pretend play ability, $B = -1.17$, $SE = 0.45$, $p = .01$. In other words, children with a sibling, younger or older, showed, on average, one more pretend play behaviour in their everyday play (as indicated by the Early Pretending Survey), compared to toddlers without a sibling, after accounting for demographic variables of age, parental education and toddlers' gender.

9. Overall discussion

The current report aimed to explore the relation between pretend play and executive functions via a parent consultation and empirical study. First, a consultation with parents of toddlers allowed an initial qualitative exploration of pretend play and its role in children's development from the perspective of the participating mothers. Secondly, a parent-reported survey provided a quantitative estimate of the relation between toddlers' pretend play ability and their executive functioning skills. Findings from both strands of this

Table 2
Descriptive statistics of key variables of interest.

	<i>n</i>	<i>M</i>	<i>SD</i>	Range
Early Pretending Survey				
Pretend Play Ability	213	12.41	3.53	0–18.00
Early Executive Functioning Questionnaire				
Cognitive Executive Functioning	213	5.25	0.55	3.32–6.42
Executive Functioning Regulation	213	4.89	0.96	2.00–7.00

Note. *N* = 213.24.

Table 3

Correlation analysis of key variables of interest.

	Toddlers' Age	Parental Education	Pretend Play Ability	CEF	RG
Toddlers' Age	-	.08	.47***	.21**	-.09
Parental Education	-	-	.08	.18*	.11
Pretend Play Ability	-	.05	-	.37***	-.04
CEF	-	.16*	.31***	-	.27***
RG	-	.11	.01	.29***	-

Note. $N = 213$. There was one missing response for parental education. Correlations were computed using Spearman's rank correlation. The lower triangle of the matrix shows partial correlations, controlling for toddlers' age. CEF = Cognitive Executive Functioning, RG = Executive Functioning Regulation.

* $p < .05$, ** $p < .01$, *** $p < .001$.

Table 4

Multiple regression analyses predicting toddlers' cognitive executive functioning and executive functioning regulation.

Predictors	<i>B</i>	<i>SE</i>	β	95% CI	<i>t</i>	<i>p</i>
Model 1: Cognitive Executive Functioning						
Toddlers' Age	0.001	0.01	.01	-.12,.15	0.18	.85
Gender: Female	0.06	0.07	-	-	0.90	.37
Parental Education	0.11*	0.05	.16*	.03,.29	2.46	.01
Pretend Play Ability	0.06***	0.01	.42***	.28,.55	6.16	< .001
Model 2: Executive Functioning Regulation						
Toddlers' Age	-0.01	0.01	-.07	-.22,.11	-0.99	.32
Gender: Female	0.17	0.13	-	-	-1.30	.19
Parental Education	0.22*	0.09	.18*	.04,.32	2.48	.01
Pretend Play Ability	-0.01	0.02	-.02	-.17,.12	-0.32	.75

Note. $n = 210$. Responses which included missing data (i.e., parental education, gender; $n = 3$) were removed during the analysis using listwise deletion. Standardised and unstandardised coefficients are shown.

* $p < .05$, *** $p < .001$.

research project converge, indicating a close connection between pretend play and cognitive executive functions in the toddlerhood period. The consulted parents highlighted the role of pretend play in enabling toddlers to learn, construct and apply emerging cognitive skills. Quantitative analysis, in turn, indicated a significant concurrent association in parent-reported indices of toddlers' pretend play ability and their cognitive executive functioning. However, no relation was identified between parent-reported indices of toddlers' pretend play ability and their executive functioning regulation.

Overall, these findings are reflective of previous theoretical propositions which identified a significant relation between children's development of pretend play and cognitive skills of executive functions and conceptualised pretend play as a fertile context for the practice and acquisition of executive function skills (e.g., Constien et al., 2026; Gleason & White, 2023; Vygotsky, 1967; Whitebread et al., 2019). Moreover, it reflects previous empirical research on the relation between pretend play and cognitive executive functioning in toddlers (Carlson et al., 2004; Rutherford & Rogers, 2003; van Reet, 2020). However, it is notable, and in contrast with previous research (Smits-van der Nat et al., 2024), that this identified relation did not extend to executive functioning regulation within the current study. In other words, toddlers who pretend more (e.g., barking like a dog, calling one's uncle on a toy phone), are also more likely to persist in a challenging puzzle and try out different and novel solutions when a piece does not fit in comparison to peers who show a smaller range of pretend play behaviours. Yet, they are equally likely to become and stay frustrated when, after repeated tries, a puzzle piece will not fit, or they are not getting a specific toy or snack they wanted.

On the one hand, this discrepancy might relate to the notion of pretend play being a purely cognitive skill (Gleason & White, 2023). Within this interpretation, it makes sense that children's ability to pretend also relates to other cognitive skills like imagination (Gleason & White, 2023), theory of mind (Lillard, 1993), language (Quinn et al., 2018), or, such as in this study, cognitive executive functioning, but not to skills that may not be purely cognitive, like regulation. Arguably, challenges encountered in pretend play, like making a distinction between an object (e.g., a block) and its' pretend attribution (e.g., a car), are conceptually similar to those in situations that require cognitive executive functions, such as learning not to touch something, they have been told not to touch (e.g., electrical socket; inhibitory control). In both scenarios, children are able to (1) create a dual representation of the focal object in their mind (i.e., car vs. block; something I want to touch vs. something I should not touch), (2) distinguish between the two, and (3) let one over the other guide their action. This representational competence is a foundational skill of children's pretend play, but also related to their cognitive executive functions, like inhibitory control, working memory and cognitive flexibility (Carlson & White, 2013; Carlson et al., 2014).

Albertson and Shore (2009), for example, sought to assess this theoretical account by introducing pre-school children (37–59 months) to both the real and the pretend attribution of a number of objects (e.g., a straw - a pen, a spoon - a toothbrush), and subsequently, testing if they could recall both attributions. While children, overall, were better at recalling the real attribution of the shown objects, the ability to recall both the real and the pretend attributions (i.e., dual representation) was significantly related to a composite executive functioning score, which was assessed via a series of lab-based tasks (Albertson & Shore, 2009). Within this

account, skills of executive functioning may be needed to engage in pretend play in the first place (i.e., to distinguish between a block and car), while at the same time pretend play may provide a context within which to practice these emerging cognitive skills (Carlson & White, 2013; Gleason & White, 2023).

On the other hand, pretend play may not merely be a cognitive exercise, but also provides frequent opportunities for emotional experiences and expressions (Smits-van der Nat et al., 2026). Children playing together with little blocks as cars may laugh out loud, if one of them makes a particular funny beeping sound, become frustrated if a parent is not paying enough attention to their play, or fight over their favourite car. Accordingly, previous research has shown children's pretend play to be related to their development of regulatory skills (Galyer & Evans, 2001; Hoffmann & Russ, 2012; Smits-van der Nat et al., 2026). In a longitudinal observational study, for instance, Lindsey and Colwell (2013) showed that the frequency of pretend play behaviours in five-year old preschoolers predicted their emotional regulation, as reported by mothers, a year later. Similarly, Goldstein and Lerner (2018) showed that an 8-week dramatic pretend play intervention, which involved playing like animals characters, or embodying emotional states, was effective in improving emotional control in 4–5 year old preschoolers.

Findings from the present study contrast with these previously reported results. On the one hand, this discrepancy may stem from the fact that this study was purposefully set within the period of toddlerhood, which has not been exhaustively explored previously. On the other hand, this discrepancy may also relate to differences in methodologies, which differ widely across studies (Smits-van der Nat et al., 2024). Within the present study, both pretend play abilities and executive function regulation were assessed independently via parent-reported measures, which differs from previous studies which assessed frequency of pretend play (e.g., Galyer & Evans, 2001; Lindsey & Colwell, 2013), or emotion regulation specifically via play observations (e.g., Galyer & Evans, 2001; Slot et al., 2017; Smits-van der Nat et al., 2026) and may explain differences between studies. Slot et al. (2017), for example, showed that children's pretend play behaviours were related to scores of emotion regulation assessed via play observations, but not to separately assessed lab-based measures (i.e., Snack and Gift Delay; Slot et al., 2017).

On a theoretical basis, pretend play and regulatory skills may further be distinguished as well. While pretend play may raise emotional experiences, the actual regulation of one's own emotions (i.e., soothing, calming down, talking through events) rarely happens within the play, but rather outside the play frame (e.g., parent taking child to the side). In fact, intense, and prolonged display of emotions (e.g., crying, screaming, etc.) may be disruptive for the play to continue and require intervention (Galyer & Evans, 2001). This is specifically relevant to toddlers, who possess limited, independent regulatory skills, and thus require additional external support from parents and caregivers. In contrast, cognitive executive functions are inherently embedded within pretend play itself, as described previously. As such, abilities of pretend play, as they were assessed within the context of the current study (Hoicka & Prouten, 2024), may not matter in regard to regulatory skills of executive functions to the same degree they do for cognitive executive functions. Further longitudinal, and experimental research, specifically within the period of toddlerhood, are needed to further explore these distinctions.

9.1. Strengths and limitations

The current findings should be interpreted within the context of the strengths and limitations within the design and scope of the present study. First, the use of online methods enabled a broad recruitment effort across Northern Ireland and the Republic of Ireland and effectively included respondents across most Irish counties (e.g., Doebel & Frank, 2023). Nonetheless, the final sample was largely homogenous in regard to gender, ethnicity, and educational status. Predominantly well-educated (i.e., 64% postgraduate education), White-Irish mothers responded to the online survey. Future research may benefit from a more individual approach to reach out to underrepresented populations (e.g., fathers, lower socioeconomic status) to better reflect the target population of toddlers and their caregivers. Secondly, the qualitative component within the presented findings provide further insights from a parental perspective to the identified association between pretend play and executive functioning, yet, given the sample size, should only be considered as illustrative rather than generalizable.

Furthermore, the cross-sectional design of the study cannot identify the degree of causality or direction within the identified association between toddlers' pretend play and executive functioning. As such, further experimental and longitudinal research is indicated. Similarly, limitations in regard to the validity of parent-reported data must be acknowledged. Given the shared-method variance, the reported associations between pretend play and indices of executive functions could have been potentially inflated as both measures are based on parental observations of their toddlers' behaviour. While the inclusion of newly developed, age-appropriate measures of both pretend play and executive functioning allowed this study to effectively address the identified gap in previous research, namely toddlerhood, it may also have introduced a degree of bias into its findings. Specifically in regard to executive functions, parent-reported questionnaires and lab-based measures of executive functioning seldom converge (e.g., O'Meagher et al., 2019; Soto et al., 2020). Moreover, potential halo effects in parental ratings may have impacted measures of executive functions and pretend play and, in turn, their identified relations. Consequently, the findings from the present studies may not readily equate to studies utilising lab-based tasks and should be replicated using additional measures.

9.2. Research implications

Given that the majority of the literature to date has focused on preschool and primary school-aged children within the study of executive functions and pretend play, the findings from the present study effectively extend the current evidence base to the under-researched period of toddlerhood. Moreover, the findings reflect previous theoretical conceptualisations that identify pretend play as a fertile context for the practice and acquisition of cognitive executive function skills in early childhood (Carlson & White, 2013;

Constien et al., 2026; Gleason & White, 2023). Previous intervention research has shown that pretend play can effectively be used as a way to strengthen children's self-regulation skills in a pre-school, or primary school contexts (e.g., Goldstein & Lerner, 2018; Moore & Russ, 2008; Muir et al., 2023). The results of the current study, in turn, suggest that pretend play could also be utilised to foster cognitive executive function skills even before children enter formal school settings. This may be particularly relevant within the context of socioeconomic disparities that have been shown to affect children's early executive functions development (Lawson et al., 2018). Pretend play, as an essentially free, naturally occurring, and engaging activity for all children may be a useful target of early interventions for clinicians, teachers and families to reduce developmental inequalities and promote school readiness of all children (Thibodeau-Nielsen, Gilpin, Palermo, et al., 2020).

10. Conclusion

The current study provides insights into the concurrent association between parent-reported indices of children's pretend play and their development of executive functions during the critical, yet historically understudied, period of toddlerhood. By utilising newly developed, developmentally appropriate measures, the results of this study demonstrated a significant, positive concurrent association between indices of parent-reported pretend play and cognitive executive functions, even after controlling for a toddler's age, gender, and parental education. These findings suggest that pretend play may serve as a developmental tool through which toddlers can begin to learn, refine and develop their cognitive abilities, such as working memory, inhibitory control and cognitive flexibility. Importantly, these findings appear to reflect more than demographic or environmental factors, indicating a meaningful developmental association between pretend play and cognitive executive functions in toddlerhood. Consequently, these findings add to previous theoretical conceptualisations of pretend play being a fertile context for the practice and acquisition of executive function skills. Future research should consider extending upon this research to explore how pretend play could be implemented across diverse populations to reduce early developmental inequalities.

CRedit authorship contribution statement

Michelle Downes: Writing – review & editing, Supervision, Conceptualization. **Tobias Constien:** Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joanna Kelly:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Funding

There is no funding associated with this manuscript.

Declaration of Competing Interest

The author declares that there is no conflict of interest associated with this manuscript

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.cogdev.2026.101759](https://doi.org/10.1016/j.cogdev.2026.101759).

Data availability

All data and material presented in this article are publicly available on this study's OSF project page (osf.io/a5473).

References

- Albertson, K., & Shore, C. (2009). Holding in mind conflicting information: Pretending, working memory, and executive control. *Journal of Cognition and Development*, 9(4), 390–410. <https://doi.org/10.1080/15248370802678240>
- Baker, J. R., Raman, S., Kohlhoff, J., George, A., Kaplun, C., Dadich, A., Best, C. T., Arora, A., Zwi, K., Schmied, V., & Eapen, V. (2019). Optimising refugee children's health/wellbeing in preparation for primary and secondary school: A qualitative inquiry. *BMC Public Health*, 19(1), 812. <https://doi.org/10.1186/s12889-019-7183-5>
- Bauer, R. H., Gilpin, A. T., & Thibodeau-Nielsen, R. B. (2021). Executive functions and imaginative play: Exploring relations with prosocial behaviors using structural equation modeling (Article) *Trends in Neuroscience and Education*, 25, Article 100165. <https://doi.org/10.1016/j.tine.2021.100165>.
- Bell, M. A. (2012). A psychological perspective on working memory performance at 8 months of age. *Child Development*, 83(1), 251–265. <https://doi.org/10.1111/j.1467-8624.2011.01684.x>
- Berk, L. E., & Meyers, A. B. (2013). The role of make-believe play in the development of executive function: Status of research and future directions. *American Journal of Play*, 6(1), 98–110.
- Bierman, K. L., & Torres, M. (2016). Promoting the development of executive functions through early education and prevention programs. In J. A. Griffin, P. McCordle, & L. S. Freund (Eds.), *Executive function in preschool-age children: Integrating measurement, neurodevelopment, and translational research* (pp. 299–326). American Psychological Association. <https://doi.org/10.1037/14797-014>.

- Braun, V., & Clarke, V. (2021). One size fits all? What counts as quality practice in (reflexive) thematic analysis? *Qualitative Research in Psychology*, 18(3), 328–352. <https://doi.org/10.1080/14780887.2020.1769238>
- Braun, V., & Clarke, V. (2022). *Thematic Analysis*. SAGE.
- Brown, J., & Isaacs, D. (2005). *The World Café: shaping our future through conversations that matter*. Berrett-Koehler.
- Byrne, R., Duncan, A., Pickar, T., Burkhardt, S., Boyd, R. N., Neel, M. L., & Maitre, N. L. (2019). Comparing parent and provider priorities in discussions of early detection and intervention for infants with and at risk of cerebral palsy. *Child: Care, Health and Development*, 45(6), 799–807. <https://doi.org/10.1111/cch.12707>
- Cabrera, N. J., Karberg, E., Malin, J. L., & Aldoney, D. (2017). The magic of play: Low-income mothers' and fathers' playfulness and children's emotion regulation and vocabulary skills. *Infant Mental Health Journal*, 38(6), 757–771. <https://doi.org/10.1002/imhj.21682>
- Carlson, S. M., Mandell, D. J., & Williams, L. (2004). Executive function and theory of mind: Stability and prediction from ages 2 to 3. *Developmental Psychology*, 40(6), 1105–1122. <https://doi.org/10.1037/0012-1649.40.6.1105>
- Carlson, S. M., & White, R. E. (2013). Executive function, pretend play, and imagination. In M. Taylor (Ed.), *The Oxford Handbook of the Development of Imagination* (pp. 161–174). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780195395761.013.0011>
- Carlson, S. M., White, R. E., & Davis-Unger, A. C. (2014). Evidence for a relation between executive function and pretense representation in preschool children. *Cognitive Development*, 29, 1–16. <https://doi.org/10.1016/j.cogdev.2013.09.001>
- Constien, T., Moran, S., Rooney, B., & Downes, M. (2026). A head taller: A meta-analysis on the relation between pretend play and executive functions in early childhood (Article) *Developmental Review*, 79, Article 101249. <https://doi.org/10.1016/j.dr.2026.101249>
- Creaigh, N., Quinn, S., & Kidd, E. (2021). Symbolic play provides a fertile context for language development. *Infancy*, 26(6), 980–1010. <https://doi.org/10.1111/inf.12422>
- Cuevas, K., Swingler, M. M., Bell, M. A., Marcovitch, S., & Calkins, S. D. (2012). Measures of frontal functioning and the emergence of inhibitory control processes at 10 months of age. *Developmental Cognitive Neuroscience*, 2(2), 235–243. <https://doi.org/10.1016/j.dcn.2012.01.002>
- Diamond, A. (2016). Why improving and assessing executive functions early in life is critical. In In. J. A. Griffin, L. S. Freund, & P. McCardle (Eds.), *Executive function in preschool-age children: Integrating measurement, neurodevelopment, and translational research* (pp. 11–43). American Psychological Association. <https://doi.org/10.1037/14797-002>
- Diamond, A., & Ling, D. S. (2019). Review of the evidence on, and fundamental questions about, efforts to improve executive functions, including working memory. In In. J. M. Novick, M. F. Bunting, M. R. Dougherty, & R. W. Engle (Eds.), *Cognitive and Working Memory Training: Perspectives from Psychology, Neuroscience, and Human Development* (pp. 143–431). Oxford University Press. <https://doi.org/10.1093/oso/9780199974467.003.0008>
- Doebel, S. (2020). Rethinking executive function and its development. *Perspectives on Psychological Science*, 15(4), 942–956. <https://doi.org/10.1177/1745691620904771>
- Doebel, S., & Frank, M. C. (2023). Broadening convenience samples to advance theoretical progress and avoid bias in developmental science. *Journal of Cognition and Development*, 25(2), 261–272. <https://doi.org/10.1080/15248372.2023.2270055>
- Doebel, S., & Lillard, A. S. (2023). How does play foster development? A new executive function perspective (Article) *Developmental Review*, 67, Article 101064. <https://doi.org/10.1016/j.dr.2022.101064>
- Dunn, J., & Wooding, C. (1977). *Play in the home and its implications*. In In. B. Tizard, & D. Harvey (Eds.), *Biology of play* (pp. 45–58). Spastics International.
- Escobar-Ruiz, V., Arias-Vázquez, P. I., Tovilla-Zárate, C. A., Doval, E., & Jané-Ballabriga, M. C. (2023). Advances and challenges in the assessment of executive functions in under 36 months: A scoping review. *Advances in Neurodevelopmental Disorders*, 8, 365–383. <https://doi.org/10.1007/s41252-023-00366-x>
- Fiske, A., & Holmboe, K. (2019). Neural substrates of early executive function development. *Developmental Review*, 52, 42–62. <https://doi.org/10.1016/j.dr.2019.100866>
- Fleer, M. (2024). Infants and toddlers in imaginary play situations: The genesis and conditions that support the development of imagination (Article) *Learning, Culture and Social Interaction*, 48, Article 100851. <https://doi.org/10.1016/j.lcsi.2024.100851>
- Flynn, M., Wynne, O., Graham, A., McCormack, T., Senior, J., McMullen, J., & Downes, M. (2025). *A focus on mind-wandering in early classroom settings: An all-island perspective*. PublicPolicy. www.publicpolicy.ie/education/a-focus-on-mind-wandering-in-early-classroom-settings-an-all-island-perspective/
- Galyer, K. T., & Evans, I. M. (2001). Pretend play and the development of emotion regulation in preschool children. *Early Child Development and Care*, 166(1), 93–108. <https://doi.org/10.1080/0300443011660108>
- Gleason, T. R., & White, R. E. (2023). Pretend play as abstraction: Implications for early development and beyond (Article) *Neuroscience & Biobehavioral Reviews*, 147, Article 105090. <https://doi.org/10.1016/j.neubiorev.2023.105090>
- Goldstein, T. R., & Lerner, M. D. (2018). Dramatic pretend play games uniquely improve emotional control in young children (Article) *Developmental Science*, 21(4), Article e12603. <https://doi.org/10.1111/desc.12603>
- Hendry, A., & Holmboe, K. (2021). Development and validation of the Early Executive Functions Questionnaire: A carer-administered measure of executive functions suitable for 9- to 30-month-olds. *Infancy*, 26(6), 932–961. <https://doi.org/10.1111/inf.12431>
- Hoffmann, J., & Russ, S. (2012). Pretend play, creativity, and emotion regulation in children. *Psychology of Aesthetics, Creativity, and the Arts*, 6(2), 175–184. <https://doi.org/10.1037/a0026299>
- Hoicka, E., & Prouten, E. (2024). The Early Pretending Survey (EPS): A reliable parent-report measure of pretense type development for 4- to 47-month-olds (Article) *Cognitive Development*, 71, Article 101483. <https://doi.org/10.1016/j.cogdev.2024.101483>
- Hughes, C., Ensor, R., Wilson, A., & Graham, A. (2009). Tracking executive function across the transition to school: A latent variable approach. *Developmental Neuropsychology*, 35(1), 20–36. <https://doi.org/10.1080/87565640903325691>
- Hughes, C., & Katus, L. (2024). Emotion regulation and executive function. In In. M. A. Bell (Ed.), *Child development at the intersection of emotion and cognition* (2nd ed (pp. 75–96). American Psychological Association. <https://doi.org/10.1037/0000406-005>
- Ibbotson, P. (2023). The development of executive function: Mechanisms of change and functional pressures. *Journal of Cognition and Development*, 24(2), 172–190. <https://doi.org/10.1080/15248372.2022.2160719>
- Kane, N. (2016). The play-learning binary: U.S. parents' perceptions on preschool play in a neoliberal age. *Children & Society*, 30(4), 290–301. <https://doi.org/10.1111/chso.12140>
- Karr, J. E., Areshenkoff, C. N., Rast, P., Hofer, S. M., Iverson, G. L., & Garcia-Barrera, M. A. (2018). The unity and diversity of executive functions: A systematic review and re-analysis of latent variable studies. *Psychological Bulletin*, 144(11), 1147–1185. <https://doi.org/10.1037/bul0000160>
- LaForett, D. R., & Mendez, J. L. (2017). Children's engagement in play at home: a parent's role in supporting play opportunities during early childhood. *Early Child Development and Care*, 187(5-6), 910–923. <https://doi.org/10.1080/03004430.2016.1223061>
- Lawson, G. M., Hook, C. J., & Farah, M. J. (2018). A meta-analysis of the relationship between socioeconomic status and executive function performance among children (Article) *Developmental Science*, 21(2), Article e12529. <https://doi.org/10.1111/desc.12529>
- Lebel, C., & Deoni, S. (2018). The development of brain white matter microstructure. *Neuroimage*, 182, 207–218. <https://doi.org/10.1016/j.neuroimage.2017.12.097>
- Lillard, A. S. (1993). Pretend play skills and the child's theory of mind. *Child Development*, 64, 348–371. <https://doi.org/10.1111/j.1467-8624.1993.tb02914.x>
- Lillard, A. S. (2010). Mother-child fantasy play. In In. A. D. Pellegrini (Ed.), *The Oxford handbook of the development of play* (pp. 284–295). Oxford University Press.
- Lillard, A. S., Lerner, M. D., Hopkins, E. J., Dore, R. A., Smith, E. D., & Palmquist, C. M. (2013). The impact of pretend play on children's development: A review of the evidence. *Psychological Bulletin*, 139(1), 1–34. <https://doi.org/10.1037/a0029321>
- Lindsey, E. W., & Colwell, M. J. (2013). Pretend and physical play: Links to preschoolers' affective social competence. *Merrill-Palmer Quarterly*, 59(3), 330–360. <https://doi.org/10.13110/merrpalmquar1982.59.3.0330>
- Löhr, K., Weinhardt, M., & Sieber, S. (2020). The "World Café" as a participatory method for collecting qualitative data. *International Journal of Qualitative Methods*, 19, Article 1609406920916976. <https://doi.org/10.1177/1609406920916976>
- Metaferia, B. K., Futo, J., & Takacs, Z. K. (2021). Parents' views on play and the goal of early childhood education in relation to children's home activity and executive functions: A cross-cultural investigation. *Frontiers in Psychology*, 12. <https://doi.org/10.3389/fpsyg.2021.646074>

- Miller, M. R., Giesbrecht, G. F., Müller, U., McInerney, R. J., & Kerns, K. A. (2012). A latent variable approach to determining the structure of executive function in preschool children. *Journal of Cognition and Development*, 13(3), 395–423. <https://doi.org/10.1080/15248372.2011.585478>
- Miyake, A., & Friedman, N. P. (2012). The Nature and Organization of Individual Differences in Executive Functions: Four General Conclusions. *Current Directions in Psychological Science*, 21(1), 8–14. <https://doi.org/10.1177/0963721411429458>
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex “frontal lobe” tasks: A latent variable analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- Moore, M., & Russ, S. W. (2008). Follow-up of a pretend play intervention: Effects on play, creativity, and emotional processes in children. *Creativity Research Journal*, 20(4), 427–436. <https://doi.org/10.1080/10400410802391892>
- Morgan, B., Fiske, A., & Holmboe, K. (2024). Early executive function development: the first three years. J. H. Grafman (Ed.), *Encyclopaedia of the Human Brain* (2nd ed.). Elsevier. <https://doi.org/10.1016/B978-0-12-820480-1.00078-4>
- Muir, R. A., Howard, S. J., & Kervin, L. (2023). Interventions and approaches targeting early self-regulation or executive functioning in preschools: A systematic review (Article) *Educational Psychology Review*, 35(1), 27. <https://doi.org/10.1007/s10648-023-09740-6>
- Nesbitt, R. J., FitzGibbon, L., & Dodd, H. F. (2025). Parents perceptions of play in school: Findings from the British children's play survey. *Psychology in the Schools*, 62(9), 2928–2941. <https://doi.org/10.1002/pits.23512>
- Nicolopoulou, A. (1993). Play, cognitive development, and the social world: Piaget, Vygotsky, and beyond. *Human Development*, 36(1), 1–23. <https://doi.org/10.1159/000277285>
- O'Meagher, S., Norris, K., Kemp, N., & Anderson, P. (2019). Examining the relationship between performance-based and questionnaire assessments of executive function in young preterm children: Implications for clinical practice. *Child Neuropsychology*, 25(7), 899–913. <https://doi.org/10.1080/09297049.2018.1531981>
- Piaget, J. (1951). *Play, dreams and imitation in childhood* (1st ed.). Routledge. <https://doi.org/10.4324/9781315009698>
- Quinn, S., Donnelly, S., & Kidd, E. (2018). The relationship between symbolic play and language acquisition: A meta-analytic review. *Developmental Review*, 49, 121–135. <https://doi.org/10.1016/j.dr.2018.05.005>
- Recchia, V., Dodaro, A., De Marco, E., & Zizza, A. (2022). A critical look to community wisdom: Applying the World Café method to health promotion and prevention. *The International Journal of Health Planning and Management*, 37(S1), 220–242. <https://doi.org/10.1002/hpm.3594>
- van Reet, J. (2020). The relation between pretence and emergent executive functioning in toddlers (Article) *Infant and Child Development*, 29, Article e2200. <https://doi.org/10.1002/icd.2200>
- Ring, E., Mhic Mhathúna, M., Moloney, M., Hayes, N., Breathnach, D., Stafford, P., Carswell, D., Keegan, S., Kelleher, C., McCafferty, D., O'Keefe, A., Leavy, A., Madden, R., & Ozonyia, M. (2016). *An Examination of concepts of School readiness among parents and educators in Ireland*. <https://arrow.tudublin.ie/cgi/viewcontent.cgi?article=1031&context=aaschssrep>
- Ring, E., & O'Sullivan, L. (2017). ‘Only four parents mentioned play’: Implications from a national school readiness study in Ireland for children's learning in the early years of primary school. *LEARN Journal of the Irish Learning Support Association*, 39, 51–59.
- Rutherford, M. D., & Rogers, S. J. (2003). Cognitive underpinnings of pretend play in autism. *Journal of Autism and Developmental Disorders*, 33(3), 289–302. <https://doi.org/10.1023/A:1024406601334>
- Schiele, H., Krummacker, S., Hoffmann, P., & Kowalski, R. (2022). The “Research World Café” as method of scientific enquiry: Combining rigor with relevance and speed. *Journal of Business Research*, 140, 280–296. <https://doi.org/10.1016/j.jbusres.2021.10.075>
- Slot, P. L., Mulder, H., Verhagen, J., & Leseman, P. P. M. (2017). Preschoolers' cognitive and emotional self-regulation in pretend play: Relations with executive functions and quality of play (Article) *Infant and Child Development*, 26(6), Article e2038. <https://doi.org/10.1002/icd.2038>
- Smits-van der Nat, M., van der Wilt, F., Meeter, M., & van der Veen, C. (2024). The value of pretend play for social competence in early childhood: A meta-analysis (Article) *Educational Psychology Review*, 36(2), 46. <https://doi.org/10.1007/s10648-024-09884-z>
- Smits-van der Nat, M., van der Wilt, F., Meeter, M., & van der Veen, C. (2026). Relation between pretend play complexity and social competence in early childhood education: A mediation model. *Journal of Research in Childhood Education*, 1–18. <https://doi.org/10.1080/02568543.2026.2625064>
- Soto, E. F., Kofler, M. J., Singh, L. J., Wells, E. L., Irwin, L. N., Groves, N. B., & Miller, C. E. (2020). Executive functioning rating scales: Ecologically valid or construct invalid? *Neuropsychology*, 34(6), 605–619. <https://doi.org/10.1037/neu0000681>
- Sutcliffe, J. A., & Kaharod, D. (2020). Re-examining play in the early childhood curriculum. In In. J. J. Mueller, & N. File (Eds.), *Curriculum in early childhood education* (2nd ed., pp. 129–143). Routledge.
- Tamis-LeMonda, S. C., Uzgiris, I. C., & Bornstein, M. H. (2002). Play in parent-child interactions. In In. M. H. Bornstein (Ed.) (2nd ed.) *Handbook of parenting* (2nd ed., 5. Practical issues in parenting, pp. 221–241). Lawrence Erlbaum Associates.
- Thibodeau-Nielsen, R. B., Gilpin, A. T., Nancarrow, A. F., Pierucci, J. M., & Brown, M. M. (2020). Fantastical pretense's effects on executive function in a diverse sample of preschoolers (Article) *Journal of Applied Developmental Psychology*, 68, Article 101137. <https://doi.org/10.1016/j.appdev.2020.101137>
- Thibodeau-Nielsen, R. B., Gilpin, A. T., Palermo, F., Nancarrow, A. F., Farrell, C. B., Turley, D., DeCaro, J. A., Lochman, J. E., & Boxmeyer, C. L. (2020). Pretend play as a protective factor for developing executive functions among children living in poverty (Article) *Cognitive Development*, 56, Article 100964. <https://doi.org/10.1016/j.cogdev.2020.100964>
- Thompson, B. N., & Goldstein, T. R. (2019). Disentangling pretend play measurement: Defining the essential elements and developmental progression of pretense. *Developmental Review*, 52, 24–41. <https://doi.org/10.1016/j.dr.2019.100867>
- Thompson, A., & Steinbeis, N. (2020). Sensitive periods in executive function development. *Current Opinion in Behavioral Science*, 36, 98–105. <https://doi.org/10.1016/j.cobeha.2020.08.001>
- United Nations. (1989). *Convention on the Rights of the Child*. In *Treaty Series*, 1577 p. 3).
- Varghese, R., Fiske, A., & Holmboe, K. (2026). Neural substrates of executive function development in children under three: A mini-review of recent advances. *Frontiers in Developmental Psychology*, 4, 2026. <https://doi.org/10.3389/fdyps.2026.1752676>
- Vygotsky, L. S. (1967). Play and its role in the mental development of the child. *Soviet Psychology*, 5(3), 6–18. <https://doi.org/10.2753/RPO1061-040505036>
- Whitebread, D., Grau, V., Kumpulainen, K., & McClelland, M. M. (2019). The importance of play, oral language and self-regulation in children's development and learning: Implications for quality in early childhood education. In D. Whitebread, V. Grau, K. Kumpulainen, M. M. McClelland, P. N. E., & D. Pino-Pasternak (Eds.), *The SAGE Handbook of Developmental Psychology and Early Childhood Education* (pp. 554–569). SAGE Publications Ltd. <https://doi.org/10.4135/9781526470393.n32>
- Wiese, S. A., Sheffield, T., Nelson, J. M., Clark, C. A. C., Chevalier, N., & Espy, K. A. (2011). The structure of executive function in 3-year-olds. *Journal of Experimental Child Psychology*, 108(3), 436–452. <https://doi.org/10.1016/j.jecp.2010.08.008>
- Wright, C. A., Pasek, J., Lee, J. Y., Masters, A. S., Golinkoff, R. M., Thomsen, B. S., & Hirsh-Pasek, K. (2023). U.S. parents' attitudes toward playful learning (Article) *Frontiers in Developmental Psychology*, 1, Article 1267169. <https://doi.org/10.3389/fdyps.2023.1267169>
- Yeung, E., Müller, U., & Carpendale, J. I. M. (2019). Developmental continuity between social-cognitive skills at age 2 and false belief understanding at age 4. *Cognitive Development*, 50, 157–166. <https://doi.org/10.1016/j.cogdev.2019.05.002>
- Zelazo, P. D., & Carlson, S. M. (2012). Hot and cool executive function in childhood and adolescence: Development and plasticity. *Child Development Perspectives*, 6(4), 354–360. <https://doi.org/10.1111/j.1750-8606.2012.00246.x>
- Zelazo, P. D., & Carlson, S. M. (2023). Reconciling the context-dependency and domain-generalty of executive function skills from a developmental systems perspective. *Journal of Cognition and Development*, 24(2), 205–222. <https://doi.org/10.1080/15248372.2022.2156515>
- Zelazo, P. D., Morris, I. F., Qu, L., & Kesek, A. C. (2024). Hot executive function: emotion and the development of cognitive control. In M. A. Bell (Ed.), *Child development at the intersection of emotion and cognition* (2nd ed (pp. 51–73). American Psychological Association. <https://doi.org/10.1037/0000406-004>